Longitudinal Data Analysis
Assignment 2: Non-Gaussian Longitudinal Data

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1 Introduction

This report looks further into the longitudinal data of a study on Alzheimer's disease. The study followed 1253 patients starting with baseline measurements at their first diagnosis and then annually over six years. Baseline patient data included in this report are age (AGE), sex (SEX), body mass index (BMI) and activities of daily living (ADL). Four variables were measured at each of the seven time points: two cognitive scales (CDR-SB and BPRS) and two possible biomarkers (ABPET and TAUPET). The first report focused on how Brief Psychiatric Rating Scale (BPRS) changed over time, while this report looks into the other cognitive scale data, Clinical Dementia Rating Sum of Boxes (CDRSB). The main research question was to examine the importance of ADL, ABPET, and TAUPET for CDRSB, while accounting for sex, age, and body mass index. Specific research questions were the following:

1. What are the effects of ADL, ABPET and TAUPET on the longitudinal profiles of CDRSB at the population level?

2. What are the effects of ADL, ABPET and TAUPET on the longitudinal profiles of CDRSB at the patient level? What can be observed from the empirical Bayes predictions of the random effects?

2 Statistical methods

2.1 Data preparation and exploration

The CDRSB data were dichotomized to be 0 if CDRSB ≤ 10 (so for less advanced dementia) and 1 if CDRSB > 10 (so more advanced dementia), referred to as low and high CDRSB, repectively, in this report.
For data visualization, the proportion of patients with high CDRSB was plotted versus years for groups of patients according to age, sex, BMI, ADL and ABPET. Age groups were defined as (1) ≤ 67, (2) $68 - 77$, and (3) ≥ 78. For BMI, common categories of normal ($18.5 \leq BMI \leq 24.9$), overweight ($25 \leq BMI \leq 29.9$) and obese ($BMI \geq 30$) were used. For ABPET, two groups were used with values (1) < 2.5, (2) ≥ 2.5. Multiple questionnaires exist that assess performance in ADLs of adults living with Alzheimer's disease and scale and scoring methods vary (Osman et al., 2025). It was not clear from the dataset which questionnaire or scale was used, so for the visualization four categories were determined ad hoc with ADL (1) between 0 and 4, (2) between 5 and 9, (3) between 10 and 14, (4) between 15 and 20. For the analyses, ADL was considered as a continuous variable.

2.2 Marginal model

A Generalized Estimating Equation (GEE) model was used to analyze the longitudinal binary outcome Y_{ij}, showing whether a patient's CDRSB score exceeded 10 at measurement j ($Y_{ij} = 1$ if CDRSB > 10, and 0 otherwise). The full model was formulated as:

$$\begin{aligned}\text{logit}\left[P(Y_{ij}=1|Year_{ij},Sex_i,Age_i,BMI_i,ADL_i,ABPET_i)\right]=\\ \beta_0+\beta_1\text{Sex}_i+\beta_2\text{Age}_i+\beta_3\text{BMI}_i+\beta_4\text{ADL}_i+\beta_5\text{ABPET}_i\\ +(\beta_6+\beta_7\text{Sex}_i+\beta_8\text{Age}_i+\beta_9\text{BMI}_i+\beta_{10}\text{ADL}_i+\beta_{11}\text{ABPET}_i)\cdot Year_{ij}\end{aligned} \tag{1}$$

where:

- $Year_{ij}$: is the time point at which the jth measurement is taken for the ith subject.
- Sex: Dummy variable with Male as reference group (0: Male, 1: Female).
- $Age_i, BMI_i, ADL_i, ABPET_i$: Baseline covariates for patient i. All these covariates were considered as continuous.
- The β coefficients represent population-average fixed effects for the corresponding covariates and their interactions with time.

Because the measurements were equally spaced, two working correlation structures (unstructured and AR(1)) were used when fitting the model because some correlation between measurements was expected, and this correlation should decrease as the time between measurements increased. Other structures, such as independence or exchangeable, were considered less appropriate for this study. Robust empirical standard errors were used for inference, and model-based standard errors were also calculated for comparison and to informally assess which working correlation fit the best.
To evaluate the importance of ADL and ABPET for the CDRSB longitudinal profiles, it was tested whether the coefficients $\beta_4, \beta_5, \beta_{10}, \beta_{11}$ were significantly different from zero. These coefficients captured the baseline effects and time-interaction effects associated with ADL and ABPET.

2.3 Random-effects model

As a subject-specific approach, a Generalized Linear Mixed Model (GLMM) was fitted to the longitudinal binary outcome Y_{ij} (see Section 2.2). A logistic model with linear time trends was assumed, including all baseline covariates for each patient. Correlation between repeated measurements within the same patient was accounted for by including random intercepts and random slopes. More specifically, the model is given by

$$\begin{aligned}\text{logit}\left[P(Y_{ij}=1|Year_{ij},Sex_i,Age_i,BMI_i,ADL_i,ABPET_i,b_{0i},b_{1i})\right]=\\ \beta_0+b_{0i}+\beta_1\text{Sex}_i+\beta_2\text{Age}_i+\beta_3\text{BMI}_i+\beta_4\text{ADL}_i+\beta_5\text{ABPET}_i\\ +(b_{1i}+\beta_6+\beta_7\text{Sex}_i+\beta_8\text{Age}_i+\beta_9\text{BMI}_i+\beta_{10}\text{ADL}_i+\beta_{11}\text{ABPET}_i)\cdot Year_{ij}\end{aligned} \tag{2}$$

where:

- b_{0i}, b_{1i} are the subject-specific random intercept, and slope, assumed to follow

$$\mathbf{b}_i=(b_{0i},b_{1i})^\top\sim N(\mathbf{0},\mathbf{D}).$$

- The β coefficients represent subject-specific fixed effects for the corresponding covariates and their interactions with linear time.
- All variables are defined as in Section 2.2.

Model estimation was performed using maximum likelihood, with the adaptive Gaussian quadrature (AGQ) method used to approximate the likelihood in the presence of random effects. Different numbers of quadrature points were examined to ensure stability and accuracy of the parameter estimates.
The empirical Bayes estimates of the random effects were also obtained and examined to assess subject-specific deviations from the population-averaged trajectories.
As in the GEE analysis, we are interested in assessing whether the baseline and time-varying effects of ADL and ABPET influence the probability of having a CDRSB score greater than 10 over time.

3 Results and discussion

3.1 Data exploration

Overall, from baseline to year 3 about 28% of patients had high CDRSB, and thus more advanced dementia (Figure 1, left). In year 4 the frequency of high CDRSB increased sharply to 44% and in years 5 and 6 it was more than 60%. The overall retention rate of patients was only 40.8% in year 6 (Table 1). About 30% of male and 25% of female patients had high CDRSB in the first three years, while in the last two years the proporion was slightly higher for female than for male patients (Figure 1, right). Retention rate was higher for male than for female patients, at 55.4% and 26.7% in year 6, respectively (Table 1). The youngest age group ($\leq$ 67 years old) had the highest proportion of patients with high CDRSB from year 1 until year 6 (Figure 2, left), which seemed surprising, but this group also had a high retention rate, unlike the oldest age group ($\geq$78 years old) which had a dropout rate of 39.5% already in the first year. The CDRSB profiles seemed similar in the three BMI groups, and the obese group was small and had the lowest retention rate so the results should be interpreted with caution (Figure 2, Table 1).

For TAUPET at baseline, 1197 patients had a value of 1.9. The 56 patients with a TAUPET value > 1.9 all dropped out of the study by year 3 (Table 1), so it was not possible to assess how the baseline TAUPET value related to the probability of high CDRSB over time. The patients with low ADL (between 0 and 4) had a low retention rate, by year 4 only two patients of this group remained in the study and these also dropped out in year 5. Almost three quarters of patients had an ADL score between 5 and 9 and this group had a retention rate of 40.7% in year 6. Of the few patients with higher ADL scores (104 with ADL between 10 and 14 and 58 with ADL between 15 and 20), all patients remained in the study until the end (Table 1). There were no clear differences in high CDRSB profiles among the ADL groups (Figure 3, left). The proportion of patients with high CDRSB was similar in both ABPET groups ovrer time, taking into account that only 28 patients with high baseline ABPET were left in the study in year 6 (Figure 3, right; Table 1).

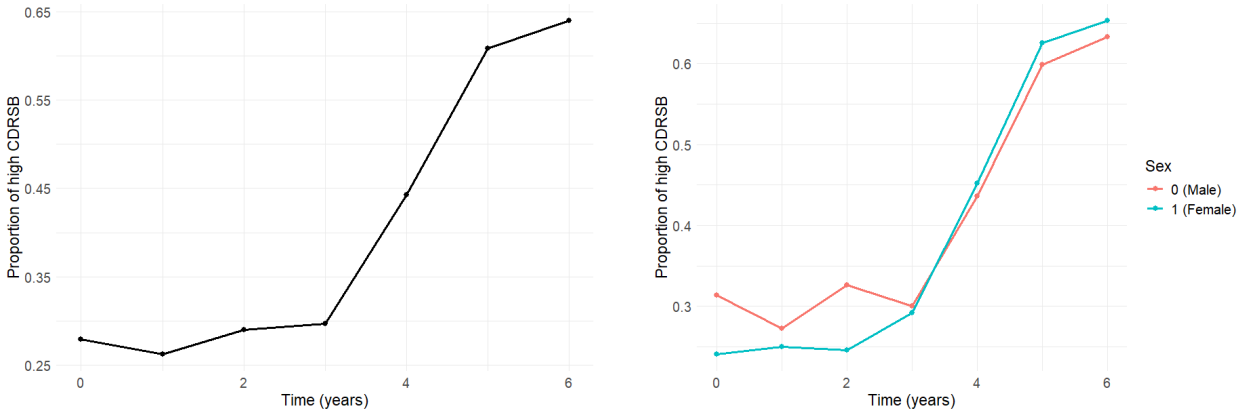


Figure 1: Proportion of high CDRSB scores over years (Left) overall (Right) for male and female patient groups.

Table 1: Retention over time by subgroups of sex, age, BMI, ADL, and baseline TAUPET, ABPET and CDRSB.

Variable	Group	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Sex	Male	616 (100.0%)	583 (94.6%)	551 (89.4%)	506 (82.1%)	463 (75.2%)	409 (66.4%)	341 (55.4%)
	Female	637 (100.0%)	523 (82.1%)	463 (72.7%)	401 (63.0%)	314 (49.3%)	243 (38.1%)	170 (26.7%)
Age group	≤67 years	318 (100.0%)	318 (100.0%)	318 (100.0%)	316 (99.4%)	312 (98.1%)	303 (95.3%)	279 (87.7%)
	68–77 years	634 (100.0%)	606 (95.6%)	571 (90.1%)	510 (80.4%)	424 (66.9%)	332 (52.4%)	223 (35.2%)
	≥78 years	301 (100.0%)	182 (60.5%)	125 (41.5%)	81 (26.9%)	41 (13.6%)	17 (5.7%)	9 (3.0%)
BMI group	Normal (18.5–24.9)	451 (100.0%)	405 (89.8%)	379 (84.0%)	346 (76.7%)	304 (67.4%)	262 (58.1%)	204 (45.2%)
	Overweight (25–29.9)	759 (100.0%)	666 (87.7%)	604 (79.6%)	534 (70.4%)	451 (59.4%)	372 (49.0%)	295 (38.9%)
	Obese (≥30)	43 (100.0%)	35 (81.4%)	31 (72.1%)	27 (62.8%)	22 (51.2%)	18 (41.9%)	12 (27.9%)
TAUPET	1.9	1197 (100.0%)	1089 (91.0%)	1008 (84.2%)	907 (75.8%)	777 (64.9%)	652 (54.5%)	511 (42.7%)
	> 1.9	56 (100.0%)	17 (30.4%)	6 (10.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
ABPET	< 2.5	845 (100.0%)	820 (97.0%)	790 (93.5%)	734 (86.9%)	668 (79.1%)	590 (69.8%)	483 (57.2%)
	≥ 2.5	408 (100.0%)	286 (70.1%)	224 (54.9%)	173 (42.4%)	109 (26.7%)	62 (15.2%)	28 (6.9%)
CDRSB	≤ 10	889 (100.0%)	797 (89.7%)	730 (82.1%)	652 (73.3%)	563 (63.3%)	461 (51.9%)	351 (39.5%)
	> 10	364 (100.0%)	309 (84.9%)	284 (78.0%)	255 (70.1%)	214 (58.8%)	191 (52.5%)	160 (44.0%)
ADL	0–4	233 (100.0%)	106 (45.5%)	46 (19.7%)	15 (6.4%)	2 (0.9%)	0 (0.0%)	0 (0.0%)
	5–9	858 (100.0%)	838 (97.7%)	806 (93.9%)	732 (85.3%)	613 (71.4%)	490 (57.1%)	349 (40.7%)
	10–14	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)
	15–20	58 (100.0%)	58 (100.0%)	58 (100.0%)	58 (100.0%)	58 (100.0%)	58 (100.0%)	58 (100.0%)
Overall		**1253 (100%)**	**1106 (88.3%)**	**1014 (81.0%)**	**907 (72.4%)**	**777 (62.0%)**	**652 (52.0%)**	**511 (40.8%)**

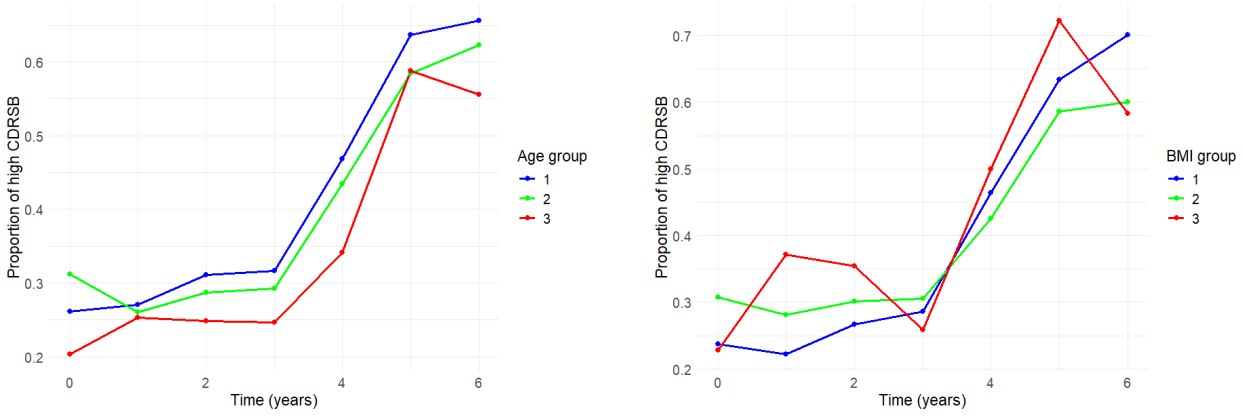


Figure 2: Proportion of high CDRSB scores over years (Left) for three age groups with (1) ≤ 67, (2) $68-77$, and (3) ≥ 78; (Right) for three BMI groups with (1) normal, (2) overweight and (3) obese.

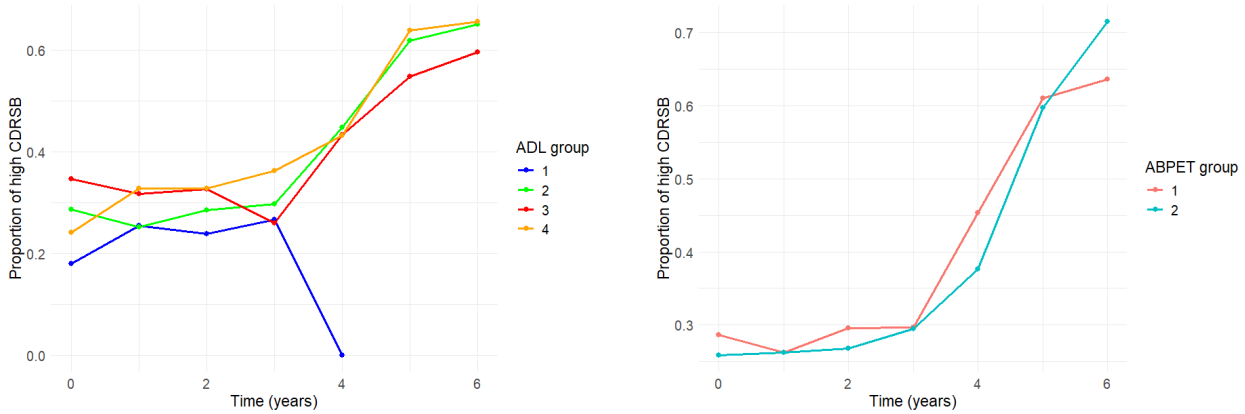


Figure 3: Proportion of high CDRSB scores over years (Left) for four ADL groups with (1) between 0 and 4, (2) between 5 and 9, (3) between 10 and 14, (4) between 15 and 20; (Right) for two ABPET groups, with (1) < 2.5, (2) ≥ 2.5.

3.2 Marginal model

Standard errors and estimated effects were similar for both the unstructured and AR(1) correlation structure, indicating that the choice of working correlation had little impact on the results, as expected (Table 2). The similarity between empirical and model-based standard errors suggested that the unstructured correlation structure better approximated the true correlation than AR(1) (Table 2). No formal inference can be made about the estimated working correlation structure from these analyses. However, unlike the AR(1) matrix, the unstructured matrix had positive, strong correlations for time points close together at the beginning and end of the study (e.g. baseline and year 1, year 5 and 6), but negative, strong correlations for time points furthest apart (e.g. baseline and year 6, year 1 and year 5) (Figure 4). The negative correlation for time points further apart could be because drop out affected the type of patient remaining in the study (those more likely to have low CDRSB) or because the patients with high baseline CDRSB remaining in the study in years 5 and 6 were those with slower progress than most.

The model accounted for sex, age and BMI. Of these three covariates, only sex and BMI were important. Holding the other covariates age, BMI, ADL and ABPET constant at baseline, the odds of female patients having a high CDRSB score were about 0.72 times the odds for male patients ($\beta_1 = -0.332$; $e^{-0.332} = 0.717$; p = 0.02). However, the difference between female and male patients decreased over years, since the sex odds ratio increased by 8.4% per year ($\beta_7 = 0.0806$; $e^{0.0806} = 1.084$; p = 0.07). Holding the other covariates age, sex, ADL and ABPET constant at baseline, the odds of having high CDRSB increased by 6.7% for each one-unit increase in BMI ($\beta_3 = 0.065$; $e^{0.065} = 1.067$; p = 0.04). Unlike the effect of sex, the effect of BMI on CDRSB decreased over time. The BMI odds ratio decreased by about 1.9% per year ($\beta_9 = -0.019$; $e^{-0.019} = 0.981$; p = 0.07), indicating that the difference in odds between patients with high and low BMI became smaller as the study progressed.

After adjusting for other covariates age, BMI, sex and ABPET, the baseline ADL score had a negligible association with the odds of having high CDRSB at baseline ($\beta_4 = 0.002$; SE = 0.029; p = 0.945). Similarly, the baseline ABPET was not associated with the odds of having high CDRSB at baseline ($\beta_5 = 0.030$; SE = 0.241; p = 0.902). The effect of ADL over time on the odds of having high CDRSB was negligible and not statistically significant ($\beta_{10} = -0.001$; SE = 0.009; p = 0.898), and the effect of ABPET over time was also not significant ($\beta_{10} = -0.009$; SE = 0.080; p = 0.908).

To get a better feel for the interpretation of the model, predicted probabilities of high CDRSB were calculated for different patients groups using the coefficient parameters estimated from the model with unstructured correlation structure for baseline and the end of the study. A common profile was selected with: Sex = 0 (male), Age = 72, BMI = 26, ADL = 6, and ABPET = 2. For this first group, the probability of having high CDRSB for these patients increased from 0.28 at baseline to 0.58 in year 6 (Table 3). This was consistent with the marginal trends observed in the exploratory analysis (Figure 1, right). Increasing ABPET from 2 to 3

(the full range present in the study) had little effect on the probability of high CDRSB. Similarly, increasing ADL from 2 to 15 while keeping other covariates constant had little effect on the predicted probability of high CDRSB at baseline or year 6 (Table 3).

Table 2: GEE parameter estimates for AR(1) and unstructured correlation models, with naive (model-based) and robust (empirical) standard errors (SE).

Effect	**Parameter**	**AR(1)**			**UN**		
		Estimate	**Naive SE**	**Robust SE**	**Estimate**	**Naive SE**	**Robust SE**
Intercept	β_0	-2.693	1.279	1.557	-2.261	1.453	1.413
Sex(Female)	β_1	-0.325	0.124	0.156	-0.332	0.140	0.141
Age	β_2	-0.005	0.016	0.020	-0.006	0.018	0.018
BMI	β_3	0.071	0.028	0.035	0.065	0.032	0.032
ADL	β_4	-0.003	0.026	0.030	0.002	0.029	0.028
ABPET	β_5	0.067	0.216	0.269	0.030	0.241	0.246
Year	β_6	1.083	0.354	0.499	0.808	0.463	0.450
Sex(Female) × Year	β_7	0.076	0.036	0.050	0.081	0.045	0.045
Age × Year	β_8	-0.003	0.004	0.006	-0.001	0.006	0.006
BMI × Year	β_9	-0.024	0.008	0.012	-0.019	0.010	0.010
ADL × Year	β_{10}	-0.0003	0.007	0.010	-0.001	0.009	0.009
ABPET × Year	β_{11}	-0.006	0.068	0.088	-0.009	0.080	0.080

AR(1)

$$\begin{bmatrix} 1.00 & 0.34 & 0.12 & 0.04 & 0.01 & 0.00 & 0.00 \\ & 1.00 & 0.34 & 0.12 & 0.04 & 0.01 & 0.00 \\ & & 1.00 & 0.34 & 0.12 & 0.04 & 0.01 \\ & & & 1.00 & 0.34 & 0.12 & 0.04 \\ & & & & 1.00 & 0.34 & 0.12 \\ & & & & & 1.00 & 0.34 \\ & & & & & & 1.00 \end{bmatrix}$$

Unstructured

$$\begin{bmatrix} 1.00 & 0.56 & 0.37 & 0.07 & -0.34 & -0.64 & -0.77 \\ & 1.00 & 0.35 & 0.12 & -0.25 & -0.52 & -0.61 \\ & & 1.00 & 0.17 & -0.03 & -0.36 & -0.43 \\ & & & 1.00 & 0.12 & -0.04 & -0.01 \\ & & & & 1.00 & 0.34 & 0.39 \\ & & & & & 1.00 & 0.62 \\ & & & & & & 1.00 \end{bmatrix}$$

Figure 4: Fitted working correlation matrices: AR(1) and unstructured.

Table 3: Predicted probabilities of high CDRSB based on the estimated model coefficients for male patients, 72 years old with BMI of 26.

Patient characteristics	**Year**	**Predicted probability**
ADL = 6, and ABPET = 2	Baseline	0.283
ADL = 6, and ABPET = 2	End of Study	0.581
ADL = 6, and ABPET = 3	Baseline	0.289
ADL = 6, and ABPET = 3	End of Study	0.574
ADL = 2, and ABPET = 2	Baseline	0.281
ADL = 2, and ABPET = 2	End of Study	0.586
ADL = 15, and ABPET = 2	Baseline	0.286
ADL = 15, and ABPET = 2	End of Study	0.569

3.3 Random-effects model

After fitting the model with 3, 5, 10 and 20 quadrature points, the model seemed to converge, with very similar parameter estimates and standard errors for 10 and 20 quadrature points (Table 4). Therefore, the estimates obtained using the adaptive Gaussian quadrature approximation with 20 quadrature points were used for reporting (Table 5).

After considering sex, age, and BMI, only sex was significantly associated with the outcome. Conditional on random intercept and slope, and adjusting for other covariates, with BMI, ADL, and ABPET held constant at baseline, female patients had lower baseline odds of having a high CDRSB score compared to male patients. Specifically, the odds for females were about 58% lower than males ($\beta_1 = -0.849$; $1-e^{-0.849} = 0.58$; $p = 0.012$). The sex odds ratio increased by nearly 27% per year (OR = 1.269), meaning the difference in odds between female and male patients became smaller over time.

Conditional on random effects and adjusting for other covariates, the baseline odds of a high CDRSB score were not significantly associated with either ADL (OR = 0.94, $p = 0.383$) or ABPET (OR = 2.22, $p = 0.171$). Baseline ADL and ABPET had no significant effect on the annual change in the odds of higher CDRSB, as indicated by the non-significant time interactions for ADL and ABPET ($p = 0.443$ and $p = 0.306$, respectively).

Table 4: Parameter estimates for the random-effects model using Adaptive Gaussian Quadrature with varying Q.

Adaptive Gaussian Quadrature					
Effect	**Parameter**	**Q=3**	**Q=5**	**Q=10**	**Q=20**
Intercept	β_0	-5.905 (3.408)	0.100 (3.551)	0.183 (3.551)	0.183 (3.553)
Sex (Female)	β_1	-0.637 (0.325)	-0.651 (0.337)	-0.846 (0.338)	-0.849 (0.338)
BMI	β_2	0.127 (0.074)	0.047 (0.077)	0.049 (0.076)	0.049 (0.076)
Age	β_3	-0.007 (0.041)	-0.069 (0.043)	-0.073 (0.043)	-0.073 (0.043)
ADL	β_4	0.018 (0.068)	-0.056 (0.070)	-0.063 (0.070)	-0.061 (0.070)
ABPET	β_5	0.222 (0.557)	0.659 (0.582)	0.802 (0.582)	0.797 (0.582)
Year	β_6	2.036 (1.053)	0.217 (1.095)	0.215 (1.095)	0.211 (1.095)
Sex(Female) × Year	β_7	0.174 (0.101)	0.178 (0.105)	0.236 (0.105)	0.237 (0.105)
BMI × Year	β_8	-0.038 (0.023)	-0.014 (0.024)	-0.015 (0.023)	-0.015 (0.023)
Age × Year	β_9	-0.005 (0.012)	0.013 (0.013)	0.014 (0.013)	0.014 (0.013)
ADL × Year	β_{10}	-0.007 (0.020)	0.015 (0.021)	0.017 (0.021)	0.016 (0.021)
ABPET × Year	β_{11}	-0.018 (0.177)	-0.146 (0.184)	-0.190 (0.184)	-0.188 (0.184)
Random Intercept Variance	d_{11}	17.327 (1.787)	18.577 (1.826)	18.711 (1.824)	18.716 (1.812)
Intercept–slope Covariance	d_{12}	-5.386 (0.541)	-5.756 (0.552)	-5.797 (0.552)	-5.797 (0.547)
Random Slope Variance	d_{22}	1.703 (0.170)	1.813 (0.174)	1.825 (0.174)	1.825 (0.173)

Table 5: Parameter estimates, standard error and P-values for the random-effects model (adaptive Gaussian quadrature with Q=20).

Effect	**Parameter**	**Estimate**	**Standard Error**	**P-value**
Intercept	β_0	0.184	3.553	0.959
Sex (Female)	β_1	-0.849	0.339	0.012
BMI	β_2	0.050	0.077	0.519
Age	β_3	-0.074	0.044	0.090
ADL	β_4	-0.062	0.071	0.383
ABPET	β_5	0.797	0.582	0.171
Year	β_6	0.212	1.095	0.847
Sex(Female) × Year	β_7	0.238	0.106	0.025
BMI × Year	β_8	-0.016	0.024	0.518
Age × Year	β_9	0.015	0.013	0.266
ADL × Year	β_{10}	0.017	0.022	0.443
ABPET × Year	β_{11}	-0.189	0.184	0.306

The covariance estimates in Table 6 describe the variability in individual baseline levels and rates of change in CDRSB. The random intercept variance was large ($d_{11} = 18.7168$), indicating substantial differences

between patients in their baseline CDRSB scores after adjusting for covariates. In contrast, the random slope variance was much smaller ($d_{22} = 1.8252$), suggesting that the rate of cognitive decline over time was relatively similar across subjects and that most heterogeneity occurred at baseline rather than in the evolution over years. The intercept–slope covariance was negative ($d_{12} = -5.7970$). In this study, most patients with high baseline CDRSB started at the top end of the scale, so they could not decline further, i.e. get a higher dementia score, whereas many patients with low baseline scores deteriorated over the course of the study. Overall, this covariance structure reflects a population in which baseline dementia severity varied widely, but yearly changes followed a more consistent pattern.

Table 6: Covariance estimates from the random effects model.

Covariance Parameter	**Estimate**	**Standard Error**
d_{11}	18.7168	1.8126
d_{12}	-5.7970	0.5479
d_{22}	1.8252	0.1732

The histogram of the empirical Bayes estimates of the random intercepts (Figure 5) shows a wide spread, indicating that patients differed considerably in their baseline CDRSB levels even after adjusting for covariates. This substantial baseline heterogeneity aligns with the large estimated intercept variance in the covariance table. In contrast, the histogram of random slopes (Figure 6) is much narrower, reflecting that patients progress at broadly similar rates over time. Only small deviations from the average yearly change were observed, consistent with the relatively small slope variance. Taken together, the two histograms suggest that most between-subject variation occurred at baseline, while the progression of CDRSB over time was more homogeneous across individuals.

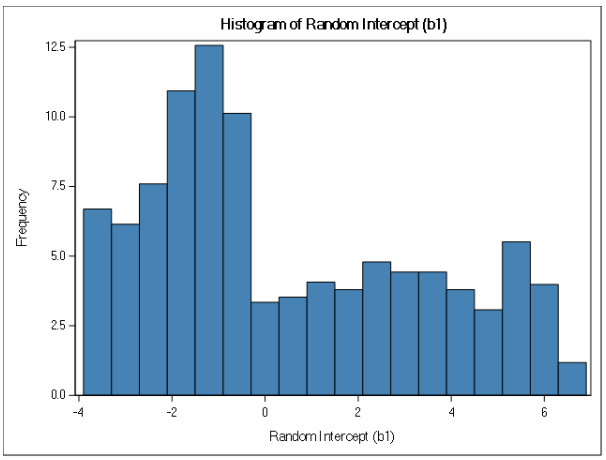


Figure 5: Histogram of empirical Bayes estimates of random intercepts (b_{1i})

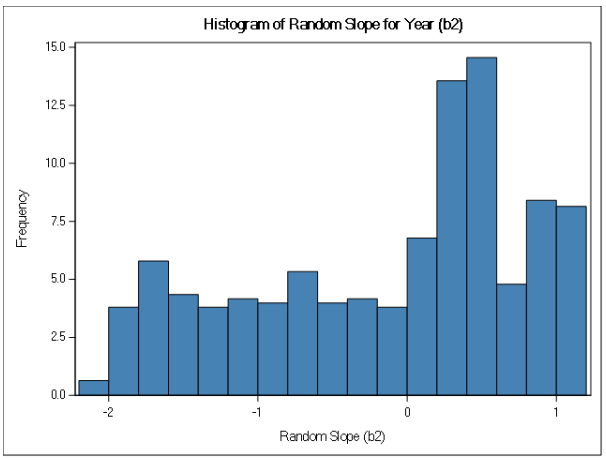


Figure 6: Histogram of empirical Bayes estimates of random slopes (b_{2i})

The points in Figure 7 form a clear downward trend, indicating a strong negative association between baseline impairment and subsequent rate of decline. Patients who began the study with higher CDRSB values tended to have smaller or slower increases over time, whereas patients with lower baseline scores tended to show faster deterioration. This visual pattern corresponds to the negative intercept–slope covariance observed in the covariance estimates (Table 6) and highlights a systematic link between where patients start and how quickly their condition progressed.

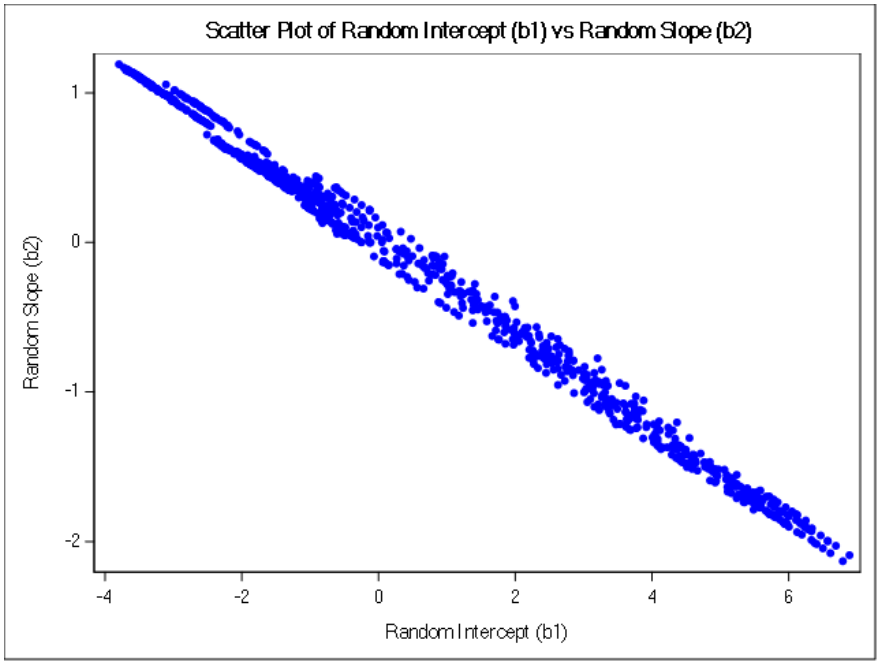


Figure 7: Scatter plot of random intercept (b1) vs random slope (b2)

3.4 Comparison of the marginal and the random-effects model

The estimates from the GEE and the GLMM differed in both magnitude and direction, and the ratio of GLMM to GEE coefficients was not constant across covariates. This behaviour was not unexpected in the setting of this study, because the two models target different estimands: GEE provides marginal (population-average) effects, whereas the GLMM yields conditional (subject-specific) effects. In the fitted GLMM, the random-intercept variance was very large ($d_{11} = 18.7$), indicating substantial heterogeneity in baseline CDRSB risk across subjects. As a result, conditional log-odds ratios tended to be much larger in magnitude than the corresponding marginal effects, leading to high ratios for covariates such as ADL and ABPET. Moreover, the random-slope variance ($d_{22} = 1.82$) and the strong negative intercept–slope covariance ($d_{12} = -5.8$) caused subject-specific time trajectories to differ markedly from the marginal trajectory. When averaging over this heterogeneous random-effects distribution, each time interaction term is distorted differently, explaining why the ratios for interaction effects varied widely and may even change sign. Overall, the pattern in Table 7 reflects that, under a logit link with large between-subject variability and correlated random slopes, GEE and GLMM coefficients cannot be interpreted as scaled versions of one another, but rather represent two distinct types of effects.

Table 7: Comparison of GEE (UN) and random-effects (GLMM) estimates, with ratios.

Effect	GEE (UN)	GLMM	Ratio (GLMM/GEE)
Intercept	-2.261	0.184	-0.08
Sex (Female)	-0.332	-0.849	2.56
Age	-0.006	-0.074	12.33
BMI	0.065	0.050	0.77
ADL	0.002	-0.062	-31.00
ABPET	0.030	0.797	26.57
Year	0.808	0.212	0.26
Sex(Female) × Year	0.081	0.238	2.94
Age × Year	-0.001	0.015	-15.00
BMI × Year	-0.019	-0.016	0.84
ADL × Year	-0.001	0.017	-17.00
ABPET × Year	-0.009	-0.189	21.00

4 Concluding remarks

Baseline ADL or ABPET did not significantly affect the odds of a patient having high CDRSB (i.e. more advanced dementia) at baseline for patients with the same sex, age, and BMI, at the marginal or patient level. Moreover, ADL and ABPET did not affect progression over years. The correction for sex was important in both the marginal and the random-effects model, with women less likely to belong to the high-CDRSB group at baseline than men with the same age, BMI, ADL and ABPET. In the marginal model and holding the other covariates constant at baseline, the odds of having high CDRSB increased with increasing BMI. In the random-effects model, the high variance of the random intercepts reflected the high baseline variability of patients' CDRSB scores, while the lower variance of the random slopes suggested more similar disease progression in individuals over time. The large variability in baseline CDRSB scores and the correlation between random intercepts and slopes resulted in a highly variable ratio of estimates of conditional (subject-specific) and marginal (population-average) effects. Estimates not being simple scaled versions of each other reflected how these models target distinct types of effects and provide complementary perspectives on disease progression.

5 References

Osman, A., Radman, D., Belchior, P., Gélinas, I., 2025. A systematic review of psychometric properties of questionnaires assessing activities of daily living among older adults with neurocognitive disorders. Aust. Occup. Ther. J., 72:e13013. DOI: 10.1111/1440-1630.13013.

6 Appendix: SAS code

```
/*--------------- Marginal model: GEE -------------------*/

/* Working correlation structures unstr was also used by adjusting the repeated statement */

proc genmod data=alzheimer25_long descending;
    class patid yearclass sex (ref='0');
    model cdrsbb = sex bmi age adl abpetbl year
                sex*year bmi*year age*year adl*year abpetbl*year
                / dist=binomial;
    repeated subject=patid / withinsubject=yearclass type=ar(1) covb corrw modelse;
run;
```

```
/*-------------- Random-effects model: GLMM -----------------*/

/* Q was adjusted in successive runs to 5, 10, 20 an 30 */

proc nlmixed data=datasets.alzheimer25_long qpoints=30;

   parms
      beta0 = 0.26
      beta_sex = -0.71
      beta_bmi = 0.05
      beta_age = -0.08
      beta_adl = -0.05
      beta_abp = 1.02
      beta_yr = 0
      beta_sex_yr = 0.19
      beta_bmi_yr = -0.013
      beta_age_yr = 0.019
      beta_adl_yr = 0.017
      beta_abp_yr = -0.26
      d11 = 18
      d12 = 4
      d22 = 1.8
   ;

   /* Linear predictor */
   teta = beta0
        + b1
        + beta_sex * sex
        + beta_bmi * bmi
        + beta_age * age
        + beta_adl * adl
        + beta_abp * abpetbl

        + b2 * year
        + beta_yr * year
        + beta_sex_yr * sexyear
        + beta_bmi_yr * bmiyear
        + beta_age_yr * ageyear
        + beta_adl_yr * adlyear
        + beta_abp_yr * abpetbly
   ;

   /* Normal likelihood for outcome */
   expteta = exp(teta);
   p = expteta/(1+expteta);
   model cdrsbb ~ binary(p);

   /* Random intercept and slope*/
   random b1 b2 ~ normal([0, 0] , [d11, d12, d22])
   subject=patid out=datasets.bayesestimates;

run;
```